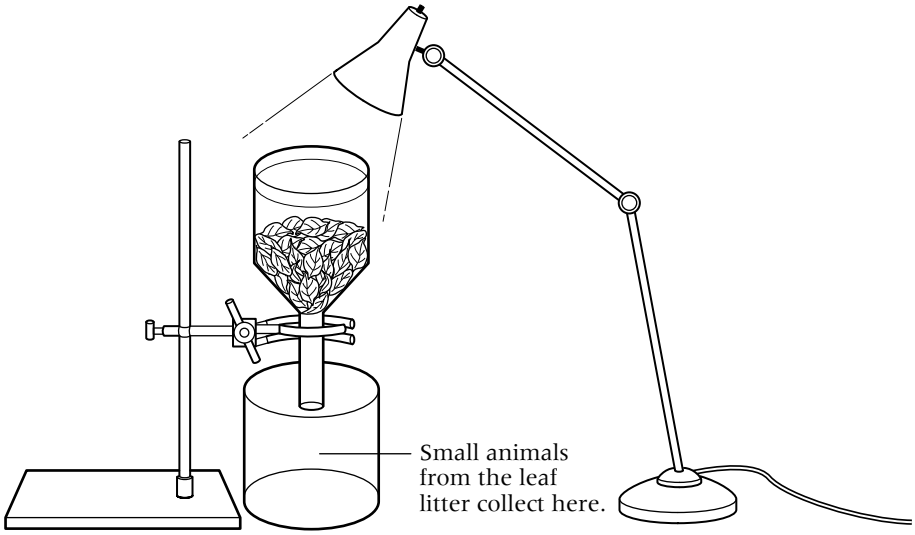
7Ce/6 Food webs and numbers

1 Leaf litter (dead leaves) was collected from a single area (0.25 m^2) in a wood. The leaf litter was placed in the apparatus shown below (called a Tullgren funnel). It was left for 24 hours. The organisms were then identified and counted. The results are shown in the table.



Name of organism	Number found
armadillo mite	102
centipede	2
ground beetle	2
hunting mite	36
spider	10

a Why do you think that the animals collect in the beaker?

b The numbers of animals get smaller as you go along a food chain. Draw a food *web* for the organisms.

c What do you think the armadillo mites were feeding on?

d What do you think would happen to the other organisms if all the hunting mites died off?

e Which two organisms are competing with each other?

f If all the leaf litter in the wood covered an area of 200 m^2, calculate how many beetles you would expect to find in the wood. Show your working.

2 a All food chains need energy. Explain how energy gets into a food chain. Use the words **producer**, **sunlight** and **chemical energy** in your answer.

b Not all the chemical energy stored in one organism can be stored in the next organism in a food chain. Suggest why not.

S knowledge, numeracy

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